This assignment contains 2 datasets. For each of the two datasets the training and testing has been carried out in the same way. I have ensured that In train_test_split the stratify parameter ensures that the training and testing sets have the same proportion of classes (or labels) as the original dataset. I have used a 5 fold CV on the training set to choose the hyperparameters and the remaining test set has been used for the confusion matrix and the ROC curve.

Part 1 - Dataset 1

Dataset 1 has 1338 points with two input features (x1,x2) and target (y). For the target we have 904 '-1' points and 434 '+1' points. This gives an indication of imbalance within the classes. The plot shows the +1 points above the curved u shaped boundary and -1 points below the boundary.

(a) Logistic regression

Logistic regression is a simple model that produces a linear boundary using the linear combination of input features and since our data is not linearly separable we use polynomial transformations so that the model can learn the curved boundary. In our case we are using a range of polynomial degrees (1-6) to see at which model is the most stable. We choose q = 2 with 5 features being x1,x2,x1x2,x1^2,x2^2 moreover an L2 regularisation has also been imposed on the model which is dependent on the value of C.

Cross-validation

For q in range (1-6) we varied the C from {0.001, 0.01, 0.1, 1, 10, 100, 1000}. We used a 5 fold CV on the training data. The training is split into five folds, the model is trained on 4 and evaluated on 5th in this way the model is evaluated for every fold once. We choose F1 for evaluation metrics rather than accuracy as accuracy alone is not reliable for imbalanced classes.
Baselines:
We also compare the models against the baseline classifier, one of them predicts only the most frequent class and the other that predicts +1 and -1 uniformly at random (random_state= 0).

From the plot it is clear that for C = 0.001 the f1 score is 0 for every q suggesting a strong penalty so much so that the model predicts -1 (majority class) for every point. The f1 increases with increase in C and then stays the same from C = 1 onwards. The model with q = 1 has an f1 score of 0.83 and is below the other q ranges because a straight line cannot fit a curved boundary. We can see that from q = 2 to q = 6 the f1 scores are between 0.94-0.95 and the error bars also overlap which indicates that the higher degree polynomial shows no further improvement. For this dataset we pick q = 2 and C = 1 as for me it's the simplest choice that stands at balance.

b) Knn

A Knn classifier was trained on the two features of the dataset1. The Knn model uses the nearest points in the dataset to make predictions making it easier for the model to form a curved boundary which is in contrast to what logistic regression does as it requires polynomial transformations for it to learn a curved shape.
I have used a similar 5 fold CV with F1 in the same way to compare the Knn classifier over a varied range of k values (1-51). We use odd values as they avoid tied votes.

We observe that for all k values the mean F1 score is between 0.93 and 0.955. It is lowest when k = 1 (0.93) where the model learns some noisy points and the highest at k = 5 (0.953). Values from k = 5 to k = 17 give almost the same F1 score ~(0.95) within one std deviation of each other. I picked k = 17 because a larger value of k gives a much more smooth boundary which is also less sensitive to the noise and also the fact that it was within the best error bands. As k becomes larger the F1 score decreases (0.93 at k = 45) as the neighbourhood becomes too wide to follow the curve..

The Knn classifier at k=17 gives similar predictions as compared to the logistic regression model with q = 2 and c = 1. Both models learn the curved boundary and make almost no mistakes with some mistakes near the boundary where the points overlap. The plots show the prediction of the model on the training set.
Below I evaluate the model on unseen test data.

c) Confusion Matrices

As mentioned earlier we had the 80:20 split. 20% of the data was kept to evaluate the model. All the classifier models are tested on the same set.
The confusion matrix gives a count of the correct and incorrect predictions on the test dataset.In the confusion matrix the rows represent the actual classes [+1,-1] and columns are the predicted classes [-1,+1] which gives us the four entries of the tables as:

/draw a confusion matrix/

From the table we observe that the Logistic regression model and the knn have the same accuracy and similar F1 scores. Knn predicts one less false positive and one more true negative than the logistic model. The baseline classifier (most frequent) predicts -1 for everything as expected with an accuracy of 67.5 % it never finds a +1 label so it's f1 score is 0. The baseline classifier (random) has 43.3% accuracy and an f1 score of 0.0315. The knn and logistic model clearly beat the baseline models.

It is also clear from the baseline model (most frequent) that accuracy alone can't determine the models performance especially when the classes are imbalanced.

d) ROC curves

The ROC curve tries every possible threshold from very strict to very relaxed and for each of the thresholds it records two numbers on the test dataset.

- **True positive rate** of the real +1 points ; for what fraction did we catch the more higher the better.
- **False positive rate**: of the real −1 points; for what fraction did we predict +1 the lower is the better.

Each threshold gives us a point on the graph which then makes a curve.In our case for the logistic regression model this score is the predicted probability and for the knn model the score is the fraction of k neighbours that are +1. The point (0,1) is a perfect classifier and the diagonal is the random classifier.

The AUC is the area under the curve a single number summarises it. 1 stands for perfect and 0.5 is random guessing. The baseline classifiers both of them output fixed labels so just a single point on the plot.

Both models have an AUC score of 0.997 and their curves can be seen overlapping in the top left corner. The baseline classifier (most freq) is at (0,0) which is again expected as it never predicts +1 and the baseline classifier (random) is at approx (0.55,0.40) close to the diagonal as expected. The curves for our trained models are above the diagonal which is an indication of their superior performance.

e) Comparison and Suggestion

Both models have a comparable f1 score close to 0.94 and both the models make almost similar predictions on the test dataset. The models have an AUC score of 0.997 with a one error difference in the confusion matrix which is negligible. This is an easier dataset both ROC curves are close to the (0,1) point. I would choose the knn model with k = 17 in comparison to the logistic regression model with q = 2 and c = 1 by a very small margin as it doesn't require any feature transformations.

Dataset 2 :

Dataset 2 has 687 points with two input features (x1,x2) and target (y). For the target we have 252 '-1' points and 435 '+1' points. This gives an indication of imbalance within the classes and from the plot we can clearly say that the +1 and -1 points are intermixed throughout the plot. There is no observable boundary between the two labels.

(a) Logistic regression

I use the same model as in Part 1 with q from 1- 6 with l2 regularisation.

Cross - Validation :

As in the above dataset I again vary q from 1 to 6 and for each q I have the C values running from [0.001, …., 100,1000]. I use the same 5 fold CV on the training set and the same F1 score metric for the +1 class similar to how I did in part 1.

Baselines:
I use the same baseline classifiers as in the previous Dataset.

From the plot it is clear that all degrees give almost the same f1 score which is close to 0.776 for C <= 1 as we increase the value of C the degrees with high values perform worse and their error bars become wider which indicates they are overfitting. I selected q = 1 and c = 1 as my values as this is a simple model and seems to have a high f1 score.
It is important to note that 0.776 is exactly the f1 obtained from predicting +1 for every point, precision is given by 435/687 ~ 0.633 with a recall of 1. The obtained F1 = 2(0.633)/1.633 ~ 0.776 so the model is not learning anything new.

b) Knn

A Knn classifier was again trained on the two features of the dataset. I use the same 5 fold CV as in the previous dataset with similar ranges.

For small k values I observe a low f1 score (0.66 for k = 1) as the model tries to learn individual noisy points. The F1 increases with increase in k and then stays almost the same from k = 25

onwards at around 0.77. I selected k = 25 as it was the first value on the plateau and with increase in k the knn learns from more neighbours and hence we see that It increasingly predicts the +1 (majority class) so we see it's F1 reach the same value of 0.776.
In this case unlike our dataset 1 the two models fail to learn the boundary. Logistic regression predicts +1 for every point and the knn model predicts -1 for only a few points.
These plots show the models performance on the training data. Below I evaluate the model on unseen test data.

c) Confusion Matrices

Similar to dataset 1 we had the 80:20 split. 20% of the data was kept to evaluate the model. All the classifier models are tested on the same set. The confusion matrix , F1 score, and accuracy are calculated exactly as in the previous dataset.

From the table we can say that both the logistic regression model and the baseline classifier model (most freq) predict +1 for everything so they have the same accuracy of 63% and F1 score of 0.773. Knn predicts -1 for 5 test points but they are all false negatives so it essentially finds no -1 points and does worse than the baseline classifier whose accuracy is about 59.4% and F1 score of 0.745. The F1 score of 0.77 looks high because +1 is the majority class. The random baseline classifier does find some negative -1 point but its accuracy and f1 are very low. None of the two models beat the baseline classifier (most freq). As in dataset 1 this also shows that f1 and accuracy should be compared to a baseline before we deduce conclusions.

d)

The Roc curves are calculated exactly in the same way as in the above dataset1. We observe that the logistic regression model has an AUC of 0.571 and knn has an AUC of 0.470. Both of the model curves are close to the diagonal which was not the case in dataset 1. The most frequent classifier is at (1,1) since it predicts +1 for every point and the random classifier is at (0.53,0.55) as expected. The curves for the models are slightly above or below the diagonal indicating the two models don't outperform the random classifier.

e) Comparison and Suggestion

The logistic regression model has the same confusion matrix as the baseline classifier (most freq) and the Knn also performs slightly worse as all of its -1 predictions are wrong. Their AUC for the test dataset is 0.571 and 0.470. This is a complex dataset as compared to dataset 1. The ROC curves are both close to the 45 degree line in contrast to dataset 1 in which the curves

were close to (0,1).The difference in AUC comes from the test split of 138 points so I cannot claim if it’s meaningful.
I would therefore not recommend either of the two trained models for this dataset as neither of them performs better than the baseline classifier (most freq). For the separation of the two labels in this dataset we would need more informative features or some stronger models that are good with scattered data. With the two given features the model fails to find any meaningful relationship within the data.